

Class12: Population Scale Analysis

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```
# Read genotype and expression data from the class website
expr <- read.table("https://bioboot.github.io/bgg213_W19/class-material/rs8067378_ENSG000000000000.txt",
                  header = TRUE)

# Check the first few rows
head(expr)
```

	sample	geno	exp
1	HG00367	A/G	28.96038
2	NA20768	A/G	20.24449
3	HG00361	A/A	31.32628
4	HG00135	A/A	34.11169
5	NA18870	G/G	18.25141
6	NA11993	A/A	32.89721

```
# Check column names
colnames(expr)
```

```
[1] "sample" "geno"   "exp"
```

```
# Count sample size for each genotype
table(expr$geno)
```

```
A/A A/G G/G
108 233 121
```

```
# Calculate median expression for each genotype
tapply(expr$exp, expr$geno, median)
```

A/A	A/G	G/G
31.24847	25.06486	20.07363

Q13. Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

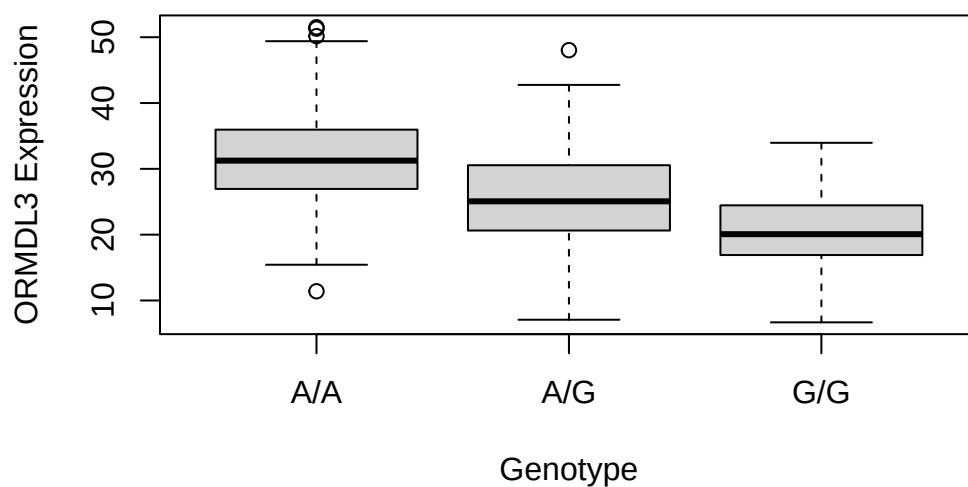
The number of samples for each genotype group was: - A/A: 108 samples - A/G: 233 samples - G/G: 121 samples

The median ORMDL3 expression values for each genotype were: - A/A: 31.24847 - A/G: 25.06486 - G/G: 20.07363

Q14. Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

```
# Make a boxplot of ORMDL3 expression by genotype
boxplot(exp ~ geno,
        data = expr,
        xlab = "Genotype",
        ylab = "ORMDL3 Expression",
        main = "ORMDL3 Expression by rs8067378 Genotype")
```

ORMDL3 Expression by rs8067378 Genotype



The boxplot shows that ORMDL3 expression differs among rs8067378 genotype groups. Individuals with the A/A genotype show the highest median ORMDL3 expression, while individuals with the G/G genotype show the lowest expression. This shows that variation at rs8067378 may influence ORMDL3 gene expression and could contribute to asthma susceptibility.